



Tree Mirroring

Let T be a rooted tree (a connected undirected acyclic graph), and let S be a perfect copy of T . Construct a new graph by taking the union of T and S , and merging the corresponding leaf nodes (but never the root). We call such a graph a *tree-mirrored graph*.

Write a program that determines if an arbitrary undirected connected graph is a tree-mirrored graph.

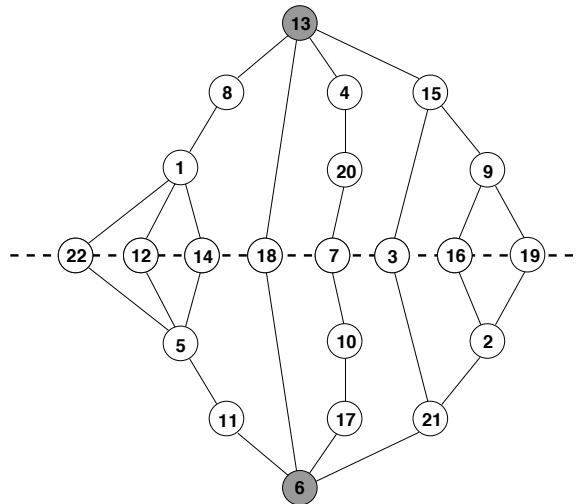


Figure 1: An example of a tree-mirrored graph. The figure corresponds to the third example test case.

Input

The first line of input contains two integers N and M , the number of vertices and edges of a graph G . The vertices in G are labeled from 1 to N . The following M lines describe the edges. Each such line contains two integers x and y ($x \neq y$; $1 \leq x, y \leq N$), describing one edge. There will be at most one edge between any pair of vertices.

Output

The first and only line of output should contain the string YES if the graph G is a tree-mirrored graph, and NO otherwise.

Constraints

$3 \leq N, M \leq 100\,000$

In test cases worth 60 points, $3 \leq N, M \leq 3\,500$. In test cases worth 30 points, $3 \leq N, M \leq 300$.

**Examples**

Input	Output
7 7 1 2 2 3 3 4 4 5 5 6 6 7 7 1	NO
6 6 1 2 2 3 2 4 3 5 4 5 5 6	YES
22 28 13 8 8 1 1 22 1 12 1 14 13 18 13 4 4 20 20 7 13 15 15 3 15 9 9 16 9 19 22 5 12 5 14 5 5 11 11 6 18 6 7 10 10 17 17 6 3 21 21 6 16 2 19 2 2 21	YES

The last example input corresponds to the graph in the figure.